

Question:

An automobile battery produces a potential difference (or "voltage") of 12 volts between its terminals. (It really consists of six 2 volt batteries following one after the other.) A headlightbulb is to be connected directly across the terminals of the battery and dissipate 40 watts of joule heat. What current will it draw and what must its resistance be?

Answer Choices:

- A. 3.0 amps, 4 ohms
- B. 2.8 amps, 3.2 ohms
- C. 3.75 amps, 3.2 ohms
- D. 4 amps, 3 ohms
- E. 3.33 amps, 3.6 ohms
- F. 5 amps, 2.4 ohms
- G. 4.5 amps, 2.67 ohms
- H. 2.2 amps, 5.45 ohms
- I. 2.5 amps, 4.8 ohms
- J. 3.5 amps, 4 ohms

Reasoning + Answer:

<think>
Use Ohm's law for a direct current circuit: the current through the bulb equals the battery's terminal voltage divided by the bulb's resistance .

- Power across the terminal pair is $P = V \cdot I$.
- Here , $P = 40 \text{ W}$ and $V = 12 \text{ V}$, so $I = P/V = 40/12 = 10/3 \approx 3.33 \text{ A}$.
- Then the bulb's resistance is $R = V/I = 12 / (10/3) = 3.6 \text{ }\Omega$.

Therefore , the correct answer is E . 3 . 33 amps , 3 . 6 oh ms .

</think>
<answer>
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</answer>